



Establishing an anthropometric database: A case for the Portuguese working population

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ABSTRACT

Up-to-date anthropometric databases can help design appropriate workstations and equipment, reducing occupational accidents and disorders. The primary aim of this study was to create a company database to support the hypothesis that the latest anthropometric database for the Portuguese working population was outdated. Through the traditional manual measurement method, this work collected data on 27 dimensions from 343 workers in an industry in Northern Portugal. The interclass correlation coefficient showed high agreeability intra- and inter-evaluators with average values of 0.940 and 0.950, respectively. Socioeconomic aspects of the sample population, including gender, immigration, and age, were evaluated. Statistical analysis included correlations between the 27 anthropometric dimensions, age, and Body Mass Index (BMI). Among the main findings, men displayed larger average dimensions than women except for hip breadth, and female body measures decreased more with age than males. Men were 1.6 times more likely to be overweight than women, whereas women were twice more likely to be obese. Concerning working shifts, individuals on the night shift showed an increased tendency to be overweight or obese (BMI > 25). Since the previous database employed a different method (photogrammetric), it was possible to compare only five dimensions (stature, knee height, elbow height, hip breadth, and abdominal depth) through mean absolute differences. Variations in the population stature and knee and elbow heights support this work's hypothesis and underscore the need to establish and update the Portuguese anthropometric database with a reliable and reproducible method.

1. Introduction

Anthropometry is the study of human body dimensions, such as stature, circumferences, and the length of segments. Anthropometric measures have direct implications in several aspects of daily life, making it possible to, for instance, evaluate children's and adolescents' nutritional status (Asif et al., 2021) as well as adults and elderly health status (Alsharif and Almuhtadi, 2021; Gába and Přidalová, 2014; Jayawardena et al., 2020). That is why several studies have been carried out across the globe to create or update anthropometric databases for general populations (Chuan et al., 2010; Klamklay et al., 2008; Parvez et al., 2022), children (Castellucci et al., 2015; Hartono, 2018; Yanto et al., 2017), the elderly and people with disabilities (Hartono, 2018; Kothiyal and Tetey, 2001). Besides these populations, anthropometric research can target working populations (Dianat et al., 2018). The latter is particularly relevant within the industrial environment since a poor observation of

anthropometric factors can result in incorrect product and workstation dimensioning, which leads to productivity loss, occupational accidents and disorders (Castellucci et al., 2020; Catapan et al., 2015). In this context, international technical standards were established to ensure the conformity of work environments with human dimensions (e.g., ISO 15535; ISO 7250–3).

Additionally, anthropometric databases have been developed to provide a reference basis to adequately dimension workstations and products (Garneau and Parkinson, 2016). Nevertheless, anthropometric databases must be regularly updated since anthropometric profiles can vary due to factors such as new nutritional habits and changes in quality of life (Galofré-Vilà, 2018; Grasgruber et al., 2016; Padez, 2002a, 2007). Examples of studies focused on working populations include Sadeghi et al. (2015), who gathered 37 anthropometric dimensions from 3720 Iranian workers, and Khadem and Islam (2014), who collected 37 anthropometric dimensions from 470 Bangladeshi male workers. Also,

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Abd Rahman et al. (2018) collected 36 anthropometric dimensions from 314 students and blue and white-collar Malaysian workers, while Sitalaksana and Widyanti (2016) collected 17 anthropometric dimensions from 660 Indonesian and Sundanese workers. All four studies collected data using traditional anthropometry equipment.

All the aforementioned works highlighted the relevance of comparing anthropometric characteristics of populations over time. Consequently, the innovation of anthropometric studies can be related to two main research challenges. First, evidencing changes in a population through anthropometric data collection, as done by Castellucci et al. (2019) in Chile when proposing an update of the past anthropometric database representing the economically active Chilean population. Second, through using new methods to collect or analyse anthropometric dimensions, such as machine learning to estimate age and sex from facial features (Porto et al., 2020). The present work is an example of the former, as it aims to obtain more recent information related to the economically active population in Portugal through a traditional data collection method, given the lack of up-to-date anthropometric data in the country, as highlighted below.

Concerning the general Portuguese population, the most recent works on anthropometric dimensions include Umami (2016) doctorate thesis, which evaluated forty dimensions related to the hands of 150 residents of North Portugal, and Fernandes (2020), who analysed the effect of furniture misfit on students' cognitive performance. However, the works of Padez (2002b) and Barroso et al. (2005) remain the most relevant and comprehensive studies on the subject due to their representativeness and robustness. Padez (2002b) evaluated the height of the Portuguese male population in military service, aged 18–24 years, which consisted of a sample of 841,457 subjects. Barroso et al. (2005) collected anthropometric measures of the Portuguese working population from 492 men and 399 women aged 17 to 65. However, despite being significant anthropometric references for the Portuguese population, these two studies, i.e., Barroso et al. (2005) and Padez (2002b), are subject to revision as they were conducted 17 and 20 years ago, respectively. As only Barroso et al. (2005) evaluated the working population from the industrial sector, it is the most recent anthropometric database of the Portuguese economically active population. Nonetheless, the authors employed a photogrammetric method instead of a traditional method (Barroso et al., 2005), which has created an obstacle for future studies when updating the existing database, as this method has not been used since then.

In relation to the working population, workers in the secondary sector (i.e., manufacturing and construction) represent approximately 25% of the economically active Portuguese population (PORDATA, 2021a). This percentage has been decreasing yearly since 1974 when it represented nearly 35% of the population (PORDATA, 2021a). This reduction may be related to industrial automation (Ramaswamy, 2018; Vermeulen et al., 2018), the ageing working population (OECD, 2019), and poor ergonomic conditions (Lucchini and London, 2014) since occupational disorders represent a global issue (WHO, 2013). According to the International Labour Organization (ILO), 340 million occupational accidents occur worldwide, with approximately 160 million people suffering from work-related disorders every year (ILO, 2020). Therefore, considering the effect of poor ergonomic conditions in the industry (Singh et al., 2022), the use of anthropometric data may help improve working conditions in the sector (Bravo et al., 2018; Castellucci et al., 2019, 2021; Viviani et al., 2018).

Based on the utility of anthropometric measures and the relevance of timely anthropometric databases to reduce occupational disorders and improve industrial environments for workers, the current study evaluates 27 anthropometric measures collected from 343 workers in a Portuguese industry. The primary aim of this study is to develop a company database and underscore the need to update the existing anthropometric database for the economically active Portuguese population, i.e., Barroso et al. (2005), as a means to allow future comparisons.

2. Materials and methods

The study was carried out in a manufacturing company in the Northern region of Portugal, whose main operations concern Mobility Solutions, Industrial Technology, Consumer Goods, and Energy and Building Technology. The sample size was established according to ISO 15535 using the following formula:

$$n = \left(1.96 \times \frac{CoV}{\alpha} \right)^2 \times 1.534^2$$

Where n represents the minimum number of subjects, CoV is the coefficient of variation obtained by dividing standard deviation (SD) by the mean, and α is the percentage of relative accuracy desired (varying from 2% to 5% (Lee et al., 2013)). The value of CoV employed in the equation was the average of the higher CoV value obtained by Barroso et al. (2005), which is body mass (15.8% for females and 14.9% for males). The α value chosen was 3%. After applying the equation, the resulting n was 237 subjects. However, the established sample population consisted of 343 workers, 174 females and 169 males, surpassing the minimum statistical sample size requirement.

Each worker that agreed to participate in this study did so voluntarily after being informed about the research purpose, methods, and the anonymity of obtained data. Furthermore, a sociodemographic questionnaire containing information about age, gender, work shift, and nationality was applied to define the social characteristics of the sample. Participants were also informed that they could leave or oppose the data collection process at any time.

2.1. Data collection

The data collection process took place from 15th June to October 14, 2021, with measures taken by three teams, each consisting of two individuals, the evaluator and the recorder. Their respective roles were to guide, accommodate, check posture, and measure subjects (evaluator), and to record data and assist the evaluator by checking the participants' posture (recorder). The measurement process consisted of checking the participant's posture, followed by the anthropometric dimension being taken and communicated by the evaluator to the recorder in a loud and clear voice. The recorder confirmed the value corresponding to the correct dimension and noted it. Data collection happened at different times to encompass different shifts, and individuals participated in the study according to the availability of production lines in the company.

Due to the SARS-COVID-19 pandemic, all individuals had their hands disinfected before and after measurement, in the same way that equipment was cleaned and disinfected frequently between participants. Using traditional methods (Fig. 1), 27 anthropometric dimensions were measured, 7 in the standing position and 20 in the sitting position (Table 1). These dimensions were selected according to workstations' characteristics in the surveyed company and the anthropometric studies of Barroso et al. (2005) and Castellucci et al. (2019). Additionally, the BMI was calculated through its formula based on body mass (BM) and stature, as shown in Table 1.

The standard procedure proposed by ISO 7250–1 was followed during the collection of the anthropometric measures, except for the body side chosen. Although ISO 7250–1 suggests the right side of the body to be the anthropometric measurement landmark, physical space and stationary equipment restrictions only allowed the evaluator to take measures from the left side of the participant subjects. However, according to Lhoman et al. (1988), the utilization of the right or the left body side has implications for skinfold measuring and not for the length of body segments. Therefore, it is considered that taking dimensions from the left body side did not compromise this work's results.

The materials used for taking measures are displayed in Fig. 1 and included: (a) a stationary anthropometer; (b) a dynamic anthropometer (Holtain Ltd®, Harpenden Anthropometer, Crymch, UK); (c) a

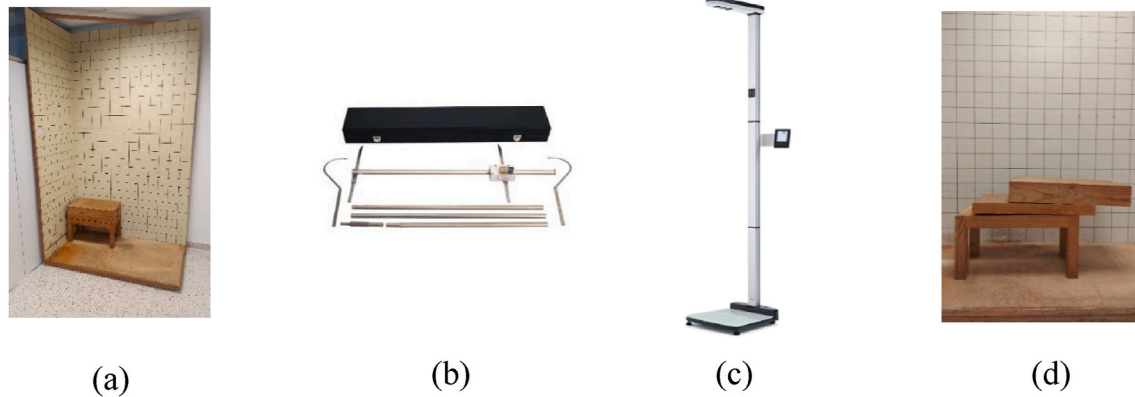


Fig. 1. Equipment used for taking anthropometric measures: (a) stationary anthropometer, (b) dynamic anthropometer, (c) measuring station of height/weight, and (d) adjustable seat.

measuring station for height and weight (SECA®, model 286); and (d) an adjustable seat with four possible height configurations (300, 350, 400 and 450 mm). The adjustable seat was used in tandem with the dynamic and stationary anthropometer to measure dimensions in sitting positions.

2.2. Intra- and inter-evaluator reliability

Three evaluators with experience in ergonomics, anthropometry, and beforehand training took the anthropometric measures. The evaluators received a two-week training on proper equipment handling and the execution of measurements by practising with 20 individuals of both genders and from varied anthropometric percentiles (i.e., 5, 50 e 95 percentiles). Individuals were measured twice by each one of the three evaluators (Table 1). As pointed out by Bravo et al. (2018), inter- and intra-evaluator reliabilities should be considered in all manuscripts to improve measurement reliability. The interclass correlation coefficient (ICC) is a type of correlation test used to measure the correlation between similar things or variables within the same class (Field, 2009). Therefore, ICC was used to assess the consistency within (intra-evaluator reliability) and between evaluators (inter-evaluator reliability) using the ‘two-way mixed’ and ‘absolute agreement’ models as recommended by Bravo et al. (2018). The range used was proposed by Portney and Watkins (2009), where values $ICC \leq 0.50$ were interpreted as weak, $0.50 \leq ICC \leq 0.75$ moderate, and $ICC \geq 0.75$ strong reliability. Table 2 illustrates that the evaluators exhibited strong inter- and intra-reliability values, except for two that presented moderate intra-evaluator reliability for shoulder breadth biacromial (intra-evaluator reliability 3) and moderate inter-evaluator reliability for foot breadth (inter-evaluator reliability 1–3). In Table 2, for instance, intra-evaluator reliability 1 shows the reliability of the measurement made by evaluator 1, whereas inter-evaluator reliability 1–2 represents the reliability of measurements between evaluator 1 and evaluator 2. The closer the value is to one, the higher the reliability of the measure taken.

2.3. Data analysis

All data collected was stored in Excel (2016) (Microsoft®). The data analysis process was performed using SPSS Statistics software version 27 (SPSS Inc., Chicago, IL, USA) and Excel 2016 (Microsoft®). The validation process consisted of three steps. First, the consistency of measures was checked through simple subtraction tests as, for instance, eye height sitting must be smaller than sitting height. Then, minimum, maximum, and mean values were employed to identify measures with ± 3 SD, as proposed in ISO 15535. Lastly, scatter plots were created to analyse the distribution of data and the presence of outliers.

After validation, data was organised and analysed. Data normality

was graphically verified using a Q-Q and P-P plots and the Skewness and Kurtosis parameters, as suggested by Field (2009) and Ruxton et al. (2015). For non-normally distributed variables, the statistical significance of using the median or the average as measures of central tendency was evaluated. To compare normally distributed measures, the average and SD were used as measures of central tendency, gender differences were calculated through the independent *t*-test (CL of 95%), and the one-sample *t*-test was used between populations (CL of 95%). In the *t*-test, *p*-values smaller than 0.05 (*p*-value < 0.05) represent a statistically significant difference between compared variables. Conversely, *p*-values greater than 0.05 (*p*-value > 0.05) represent equality between compared data (Field, 2009).

Finally, Pearson’s correlation coefficient was used to establish linear correlations between variables. The coefficients of correlations can vary between +1 and –1. A coefficient of +1 means a perfect positive relationship, 0 means no linear relationship, and –1 indicates a perfect negative relationship (Field, 2009). In addition, the magnitude of the effect can be understood as ± 0.1 representing a small effect, ± 0.3 a medium effect, and ± 0.5 a large effect (Field, 2009).

3. Results

Considering the 343 subjects participating in the sample, 169 were male (49.3%), and 174 were female (50.7%), with an average male age of 36.2 ± 9.9 years old and an average female age of 42.7 ± 11.0 years old, which matches the economically active workforce’s age range. Regarding the composition of the workforce sample, 98.5% were Portuguese workers, and 1.5% were foreign workers. Forty per cent of foreign workers consisted of Brazilians followed by Angolans (20.0%), Nepalese (20.0%), and Venezuelans (20.0%). Concerning the working shifts, 93.1% of females and 88.2% of males worked during the daytime (8 a.m.–6 p.m.), while 6.9% of the females and 11.8% of the males worked during the nocturnal period (midnight to 8 a.m.). The average body mass was 77.9 ± 11.1 kg for males and 63.7 ± 10.1 kg for females. The average stature of males and females was 1740.7 ± 66.2 mm and 1594.7 ± 62.5 mm, respectively. The average BMI was 25.7 ± 3.1 for males and 25.2 ± 4.2 for females.

Average values of body mass and measured anthropometric dimensions are shown in Fig. 2. Table 3 shows the anthropometric data for the female and male populations, including average values, SD, inter-quartile range (IR), fifth percentile (P5), fifty percentile (P50) and ninety-fifth percentile (P95). In relation to data normality, which was graphically verified through the Q-Q and P-P plots and the Skewness and Kurtosis parameters, all dimensions presented a normal distribution behaviour, except for elbow grip length for men. This dimension did not show statistically significant differences (*p*-value < 0.05) in using the average instead of the median. Therefore, the average was used as a

Table 1
Description of the evaluated anthropometric dimensions.

Equipment ^a	Dimension	Abbreviation	Definition ^b
Standing			
SHW	Body mass	BM	Total mass (weight) of the body.
SHW	Stature	Stature	Vertical distance between the floor and the top of the head measured with the subject erect and looking straight ahead (Frankfort plane).
SA	Eye height	EyeH	Vertical distance from the floor to the outer corner of the eye
SA	Shoulder height	ShoulderH	Vertical distance from the floor to the acromion
SA	Elbow height	ElbowH	Vertical distance from the floor to the lowest bony point of the bent elbow
SA	Knuckle height	KnuckleH	Vertical distance from the floor to metacarpal III
SA	Grip reach; forward reach	GRFR	Horizontal distance from a vertical surface to the grip axis of the hand while the subject leans both shoulder blades against the vertical surface
Sitting			
SA + AS	Sitting height	SittingH	Vertical distance between subject's seated surface to the top of the head measured with the subject erect and looking straight ahead (Frankfort plane)
SA + AS	Eye height sitting	EyeHS	Vertical distance between the sitting surface to the inner canthus (corner) of the eye (head in Frankfurt plane)
SA + AS	Shoulder height sitting	ShoulderHS	Vertical distance from subject's seated surface to the acromion
SA + AS	Elbow height sitting	ElbowHS	Taken with a 90° angle elbow flexion. Vertical distance from the bottom of the tip of the elbow (olecranon) to the subject's seated surface
SA + AS	Elbow grip length	EGL	Horizontal distance from back of the upper arm (at the elbow) to grip axis, with elbow bent at right angles
DA + AS	Abdominal depth	Abdominal	Maximum horizontal distance from the vertical reference plane to the front of the abdomen in the standard sitting position
DA + AS	Hand length	HandL	Perpendicular distance from a line drawn between the styloid processes to the tip of the middle finger
DA + AS	Hand breadth (metacarpal)	HandB	Projected distance between radial and ulnar at the level of the metacarpal-phalangeal joint of digit 1 to the ulnar side of the hand second to the fifth metacarpal
DA + AS	Foot breadth	FootB	Maximum distance between medial and lateral surfaces of the foot perpendicular to the longitudinal axis of the foot
DA + AS	Foot length	FootL	Maximum distance from rear of the heel to tip of the longest (first or second) toe,

Table 1 (continued)

Equipment ^a	Dimension	Abbreviation	Definition ^b
DA + AS	Buttock-knee length	BKL	measured parallel to the longitudinal axis of the foot Horizontal distance from the foremost point of the kneecap to the rearmost point of the buttock
DA + AS	Knee height	KneeH	Vertical distance from the floor or footrest to the highest point of the superior border of the patella
DA + AS	Buttock-popliteal length	BPL	Horizontal distance from the popliteal surface to the rearmost point of the buttock
DA + AS	Popliteal height	PoplitealH	Vertical distance from the floor or footrest to the posterior surface of the knee (popliteal surface)
DA + AS	Thigh clearance	ThighC	Vertical distance from the highest uncompressed point of thigh to the subject's seated surface
DA + AS	Shoulder (bideloid) breadth	SBD	Distance across the maximum lateral protrusions of the right and left deltoid muscles
DA + AS	Elbow to elbow breadth	ETEB	Maximum horizontal breadth across the elbows
DA + AS	Shoulder breadth (biacromial)	SBA	Distance between the outermost bony points on the top of each shoulder
DA + AS	Subscapular height	SSH	Vertical distance from the lowest point (inferior angle) of the scapula to the subject's seated surface
DA + AS	Hip breadth	HipB	Horizontal distance measured in the widest point of the hip in the sitting position
Calculated			
Calculated through a formula	Body Mass Index	BMI	Calculated using the formula: $BMI (kg/m^2) = Weight (kg) / Stature^3 (m^2)$.

^a Measuring station for height and weight (SHW), Stationary anthropometer (SA), Dynamic anthropometer (DA), Adjustable seat (AS).

^b Source: (ISO 7250).

measure of central tendency for all dimensions.

Relating to the CoV, 63.0% of females and 77.8% of males were within the range proposed by Pheasant (2003), as shown in Table 4. Postural factors and sample size can explain those dimensions outside Pheasant's (2003) range.

4. Discussion

4.1. Socioeconomics aspects

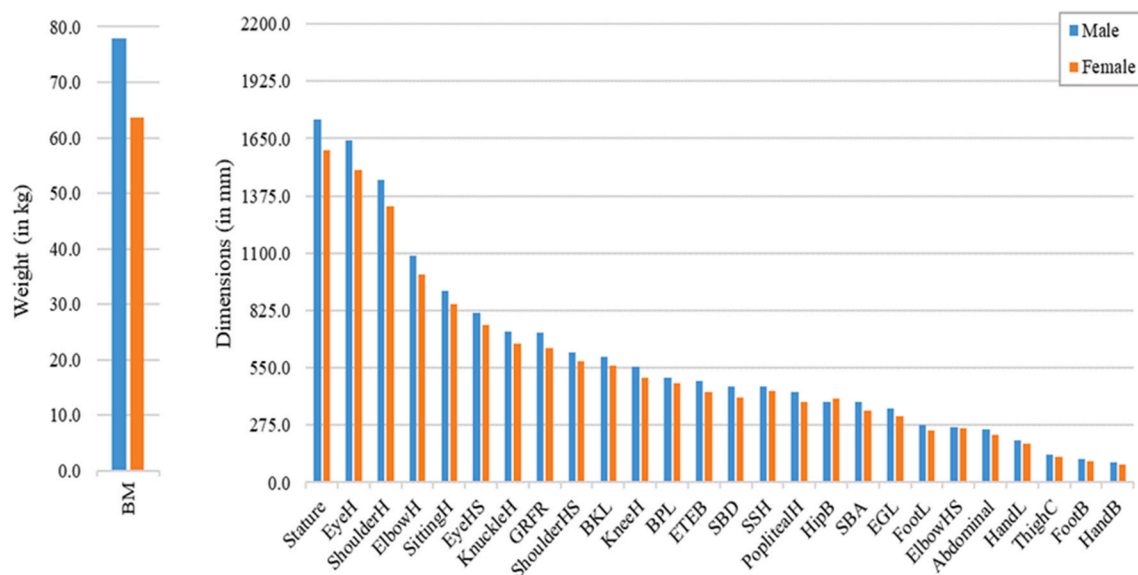
This research included subjects from a company in Northern Portugal, encompassing a working population that performed different jobs. This study's sample represented well the company's population in terms of both statistical significance ($n = 343$ in this study v. $N = 3600$ in the company; $\alpha = 3\%$) and gender proportion (females, 50.7% in this study v. 45.0% in the company; males, 49.3% in this study v. 55.0% in the company). In general, men displayed higher average body segmental dimensions than women, except for hip breadth (Fig. 2), which was also observed in other studies such as Castellucci et al. (2019) and Molenbroek et al. (2017). Also, the male (49.3%) and female (50.7%) populations were similar to the national labour force participation of 49.3% females and 50.7% males (PORDATA, 2021b). The gender employment

Table 2

ICC values indicating intra and inter-evaluator reliability.

Anthropometric Dimension ^a	Intra-evaluator reliability 1	Intra-evaluator reliability 2	Intra-evaluator reliability 3	Inter-evaluator reliability 1-2	Inter-evaluator reliability 1-3	Inter-evaluator reliability 2-3
3 EyeH	0.997	0.979	0.997	0.986	0.997	0.985
4 ShoulderH	0.996	0.995	0.995	0.996	0.996	0.997
5 ElbowH	0.989	0.984	0.983	0.985	0.983	0.981
6 KnuckleH	0.990	0.968	0.976	0.986	0.988	0.992
7 GRFR	0.949	0.958	0.903	0.964	0.948	0.935
8 SittingH	0.994	0.976	0.963	0.991	0.996	0.993
9 EyeHS	0.992	0.941	0.991	0.967	0.994	0.973
10 ShoulderHS	0.975	0.928	0.976	0.984	0.982	0.972
11 ElbowHS	0.965	0.852	0.964	0.956	0.954	0.934
12 BKL	0.902	0.755	0.966	0.955	0.929	0.964
13 KneeH	0.982	0.979	0.990	0.973	0.987	0.992
14 BPL	0.852	0.801	0.786	0.886	0.765	0.834
15 PoplitealH	0.979	0.970	0.968	0.963	0.987	0.970
16 ThighC	0.923	0.915	0.894	0.914	0.924	0.953
17 EGL	0.930	0.921	0.937	0.880	0.905	0.945
18 Abdominal	0.987	0.975	0.961	0.984	0.974	0.986
19 HandL	0.964	0.970	0.981	0.981	0.974	0.973
20 HandB	0.919	0.943	0.955	0.975	0.975	0.974
21 FootB	0.766	0.976	0.942	0.733	0.751	0.964
22 FootL	0.971	0.998	0.987	0.976	0.981	0.995
23 ETEB	0.983	0.961	0.973	0.973	0.968	0.951
24 SBD	0.993	0.974	0.976	0.984	0.989	0.964
25 SBA	0.866	0.769	0.670	0.901	0.812	0.821
26 SSH	0.912	0.863	0.768	0.875	0.872	0.833
27 HipB	0.935	0.875	0.901	0.959	0.940	0.950

^a Eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), knuckle height (KnuckleH), grip reach forward reach (GRFR), sitting height (SittingH), eye height sitting (EyeHS), shoulder height sitting (ShoulderHS), elbow height sitting (ElbowHS), elbow grip length (EGL), abdominal depth (Abdominal), hand length (HandL), hand breadth metacarpal (HandB), foot breadth (FootB), foot length (FootL), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), popliteal height (PoplitealH), thigh clearance (ThighC), shoulder bideltoid breadth (SBD), elbow to elbow breadth (ETEB), shoulder breadth biacromial (SBA), subscapular height (SSH), hip breadth (HipB).

**Fig. 2.** Average values for BM and anthropometric dimensions of male and female populations.

*Body mass (BM), stature (Stature), eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), sitting height (SittingH), eye height sitting (EyeHS), knuckle height (KnuckleH), grip reach forward reach (GRFR), shoulder height sitting (ShoulderHS), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), elbow to elbow breadth (ETEB), shoulder bideltoid breadth (SBD), subscapular height (SSH), popliteal height (PoplitealH), hip breadth (HipB), shoulder breadth biacromial (SBA), elbow grip length (EGL), foot length (FootL), elbow height sitting (ElbowHS), abdominal depth (Abdominal), hand length (HandL), thigh clearance (ThighC), foot breadth (FootB) and hand breadth metacarpal (HandB).

gap is the indicator used to show the differences between the number of men and women in the labour market. In Portugal, the employment gap was around 7.5 percentage points in 2019 (EUROSTAT, 2019). This number was below the European Union's (EU) average of 11.5 percentage points. Among the EU Members States, the gender employment gap was lowest in Lithuania (1.0 percentage points), Finland (3.5),

Sweden (4.0) and Latvia (4.3), and the largest in Malta (24.1), Italy (19.8), and Greece (19.7) (EUROSTAT, 2019).

Besides gender, immigration is another important aspect of the labour market of several countries (Castellucci et al., 2019; Gnanewarun and Bishu, 2011). In recent years, Portugal experienced an increase in the number of foreigners living in the country, surpassing the figure of

Table 3Anthropometric data collected in the present study. BM in kg; BMI in kg/m²; other measurements in mm.

Anthropometric Dimensions	Female						Male					
	Average	SD	IR	P5	P50	P95	Average	SD	IR	P5	P50	P95
1 BM	63.7	10.2	13.4	49.6	62.4	82.9	77.9	11.1	14.4	63.6	76.4	96.9
2 Stature	1594.7	62.5	87.0	1492.7	1591.5	1687.5	1740.7	66.2	87.5	1642.0	1737.0	1846.0
3 EyeH	1498.8	61.2	90.0	1393.7	1492.5	1595.0	1642.0	66.6	90.0	1542.5	1635.0	1747.5
4 ShoulderH	1325.2	58.7	80.0	1225.0	1320.0	1431.2	1452.4	59.9	92.5	1357.2	1450.0	1545.0
5 ElbowH	996.4	46.2	60.0	918.7	997.5	1080.0	1089.8	50.3	62.5	1010.0	1085.0	1180.0
6 KnuckleH	667.8	36.9	55.0	608.7	670.0	730.0	721.8	36.6	55.0	665.0	720.0	780.0
7 GRFR	646.3	29.8	40.0	595.0	645.0	696.5	716.9	36.0	55.0	655.0	715.0	782.7
8 SittingH	856.0	31.6	45.0	800.0	855.0	905.0	921.8	33.3	45.0	865.0	920.0	975.0
9 EyeHS	755.4	28.3	35.0	708.7	755.0	800.0	812.8	33.2	47.5	755.0	814.0	870.0
10 ShoulderHS	581.7	29.9	35.0	538.7	580.0	626.2	626.3	27.8	40.0	580.0	625.0	670.0
11 ElbowHS	258.5	24.5	35.0	220.0	260.0	296.2	266.7	26.3	30.0	222.2	270.0	312.7
12 BKL	562.0	26.0	35.0	520.0	560.0	600.0	601.0	27.6	40.0	560.0	600.0	645.0
13 KneeH	502.8	25.5	30.0	460.0	500.0	545.0	553.4	26.7	40.0	517.5	550.0	600.0
14 BPL	475.7	22.4	30.0	440.0	475.0	520.0	504.3	24.3	32.5	460.0	500.0	550.0
15 PoplitealH	387.9	24.7	30.0	350.0	390.0	430.0	432.1	24.3	37.5	395.0	435.0	472.7
16 ThighC	125.2	11.9	14.0	107.7	124.0	148.0	131.9	10.7	13.0	115.0	131.0	151.0
17 EGL	319.1	16.0	20.0	290.7	319.0	345.3	356.3	36.7	27.5	328.0	354.0	387.5
18 Abdominal	230.8	35.8	50.0	180.7	223.5	298.0	253.8	29.9	40.5	213.5	248.0	308.5
19 HandL	186.5	8.0	10.0	173.0	187.0	200.0	203.4	9.2	13.0	187.0	203.0	217.0
20 HandB	88.9	4.2	6.0	82.0	89.0	95.0	98.2	4.7	6.0	91.5	98.0	106.0
21 FootB	103.4	5.2	7.0	94.7	104.0	112.0	111.2	7.2	8.0	102.0	111.0	120.5
22 FootL	248.4	10.9	16.0	228.7	248.0	267.0	274.1	11.7	16.0	255.0	273.0	293.5
23 ETEB	436.8	44.3	64.0	369.7	433.5	530.7	488.6	39.8	53.5	426.5	485.0	555.5
24 SBD	406.9	18.5	23.0	380.0	405.0	441.5	463.3	23.6	29.0	427.0	463.0	512.1
25 SBA	342.8	26.4	41.0	297.7	346.0	383.2	385.5	30.2	44.5	328.5	386.0	430.0
26 SSH	438.4	26.3	36.0	394.7	437.5	485.0	458.3	27.8	42.0	411.0	460.0	499.0
27 HipB	400.9	31.1	43.0	350.7	396.5	456.5	386.1	25.8	31.0	353.5	382.0	433.0
28 BMI	25.2	4.2	5.6	19.5	24.2	34.0	25.7	3.1	4.3	21.2	25.5	31.4

*Body mass (BM), stature (Stature), eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), knuckle height (KnuckleH), grip reach forward reach (GRFR), sitting height (SittingH), eye height sitting (EyeHS), shoulder height sitting (ShoulderHS), elbow height sitting (ElbowHS), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), popliteal height (Popliteal H), thigh clearance (ThighC), elbow grip length (EGL), abdominal depth (Abdominal), hand length (HandL), hand breadth metacarpal (HandB), foot breadth (FootB), foot length (FootL), elbow to elbow breadth (ETEB), shoulder bideltoid breadth (SBD), shoulder breadth biacromial (SBA), subscapular height (SSH), hip breadth (HipB) and body mass index (BMI).

662,000 foreigners in 2020 (SEF, 2020). According to the most recent data on the occupation of the Portuguese labour market by foreigners, there were approximately 164,400 people engaged in some paid activity in 2017, which represented 3.1% of the working population at the national level (GEP, 2018). This study presented a quite similar percentage of foreign labour, as 3.4% of the investigated sample was composed of individuals from other nationalities. In case the participation of foreigners in the national workforce increases, the adjustability of working stations to different populations' anthropometric features will become essential for designing healthy and productive work environments.

One of the factors related to immigration is an ageing native population and the search for better living standards on the part of foreigners (Fougère et al., 2004). As the elderly population has been steadily increasing (Kenny et al., 2008; Shahida et al., 2015), the composition of the working population has changed as more and more elderly workers, i.e., aged over 60 years (Dyussenbayev, 2017), choose to remain active in the workforce. As reported by the World Health Organization (WHO), between 2015 and 2050, the proportion of the world population over 60 years will nearly double, going from 12 to 22% (WHO, 2021). In Portugal, the number of people in the working age (15–65 years old) is expected to decrease from 6.6 million in 2018 to 6 million in 2034, 5 million in 2054, and 4.2 million by 2080 (INE, 2020). In this study, 42.0% ($n = 143$) of the participants were 16–35 years old, 22.0% ($n = 75$) were 36–45 years old, and 36.0% ($n = 125$) were 46–65 years old. According to the classification presented by Dyussenbayev (2017), 30.6% of the participants were 'middle age' (44–60 years old), and 5.8% were 'elderly' (60–75 years old). The information related to the population age is displayed in Table 5.

4.2. Correlations between age, BMI, and anthropometric dimensions

4.2.1. Age

Age is one of the factors that imply changes in physical function (Okabe et al., 2021) and anthropometric dimensions (Cole, 2003; Hoca and Türker, 2021). As 36.4% of the individuals were found to be in the 46–65-year-old age group, it is reasonable to correlate body dimensions with age. Accordingly, certain body dimensions can decrease because of drying and reduction of the intervertebral spaces caused by ageing processes (Quittner et al., 2018). When evaluating dimensions in relation to age, it was observed that 58.6% of the correlations were significant at 0.01 ($p\text{-value} \leq 0.01$), 6.9% were significant at 0.05 ($0.01 < p\text{-value} \leq 0.05$), and 34.5% were non-significant ($p\text{-value} > 0.05$) within the female population. On the other hand, for males, 17.2% of the values were significant at 0.01 ($p\text{-value} \leq 0.01$), 10.0% were significant at 0.05 ($0.01 < p\text{-value} \leq 0.05$), and 69.0% were non-significant ($p\text{-value} > 0.05$). According to these findings, female dimensions varied more significantly with age than male dimensions (65.5% v. 31.0%).

Some dimensions had a significant negative correlation with age, but this finding was more expressive in females. In the male group, none of the dimensions showed this behaviour, against 10 dimensions presenting significant, small to medium correlations, in the female group (stature ($-0.313^{0.000}$), eye height ($-0.260^{0.001}$), shoulder height ($-0.241^{0.001}$), elbow height ($-0.246^{0.001}$), sitting height ($-0.316^{0.000}$), eye height sitting ($-0.237^{0.002}$), knee height ($-0.225^{0.003}$), buttock-popliteal length ($-0.152^{0.046}$), popliteal height ($-0.264^{0.000}$), and subscapular height ($-0.260^{0.001}$)). Seven of these dimensions are related to the loss in trunk length, while knee height and popliteal height relate to a shorter knee-floor distance, and buttock-popliteal length relates to the shortening of the gluteal-femoral segment.

Even though these reductions, especially those with a lower significance level found in the male group, can be related to the sample size

Table 4

Coefficient of Variation (%) in the Present study in reference to Pheasant (2003).

Anthropometric Dimensions	Current study		Pheasant (2003)
	Female	Male	General
1 Body mass	15.9	14.0	10–21
2 Stature	4.0	3.7	3–4
3 EyeH	4.1	4.0	3–5
4 ShoulderH	4.5	4.1	3–5
5 ElbowH	4.7	4.5	3–5
7 GRFR	4.6	5.0	4–11
8 SittingH	3.7	3.5	3–5
9 EyeHS	3.8	4.1	4–11
10 ShoulderHS	5.1	4.4	4–11
11 ElbowHS	9.5	9.7	4–11
12 BKL	4.6	4.6	4–11
13 KneeH	5.1	4.7	3–5
14 BPL	4.7	4.8	4–11
15 PoplitealH	6.5	5.6	3–5
16 ThighC	9.5	8.1	5–9
17 EGL	5.0	10.1	4–11
18 Abdominal	15.6	11.9	5–9
19 HandL	4.3	4.5	5–9
20 HandB	4.7	4.8	5–9
21 FootB	5.0	6.3	5–9
22 FootL	4.4	4.2	5–9
23 ETEB	10.5	8.1	5–9
24 SBD	4.6	5.0	5–9
25 SBA	7.8	8.0	5–9
26 SSH	6.0	5.9	3–5
27 HipB	7.7	6.6	5–9

*Body mass (BM), stature (Stature), eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), grip reach forward reach (GRFR), sitting height (SittingH), eye height sitting (EyeHS), shoulder height sitting (ShoulderHS), elbow height sitting (ElbowHS), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), popliteal height (Popliteal H), thigh clearance (ThighC), elbow grip length (EGL), abdominal depth (Abdominal), hand length (HandL), hand breadth metacarpal (HandB), foot breadth (FootB), foot length (FootL), elbow to elbow breadth (ETEB), shoulder bideltoid breadth (SBD), shoulder breadth biacromial (SBA), subscapular height (SSH), and hip breadth (HipB).

Table 5

Number of individuals by age groups.

Gender	Age groups (years old)					Total
	16–25	26–35	36–45	46–55	56–65	
Female	16	34	31	80	13	174
Male	14	79	44	25	7	169
Total	30	113	75	105	20	343

(Barroso et al., 2005), there are clinical and medical factors associated with these correlations. The larger number of variables presenting negative correlations with age in the female group can be explained by a higher average age (Female = 42.7 ± 11.0 years old v. Male = 36.2 ± 9.9 years old) and more accelerated degenerative processes in women (Curtis et al., 2015; Lang, 2011). Therefore, reductions in the female group can be explained by age-related degenerative changes that compromise musculoskeletal systems, mainly knees, hands, hips and spine, and diminish joint function (Dequeker et al., 2003; Martin and Buckwalter, 2002; Zupan et al., 2021).

The reduction in trunk length can be related to the drying and reduction of the intervertebral spaces caused by ageing processes (Quittner et al., 2018; Zupan et al., 2021). Likewise, the decrease in the knee-floor distance could be explained by the degenerative changes of the femoral-tibial and talotibial cartilages (Hudelmaier et al., 2001; Zupan et al., 2021), whereas the reduction in buttock-popliteal length segment could be explained by the deterioration in the acetabulum and triradiate cartilages (Kurth, 2021; Zupan et al., 2021).

Concerning the dimensions that showed a positive correlation with

age, it was observed that 6 of them presented small to medium correlations for females (body mass ($0.244^{0.001}$), abdominal depth ($0.465^{0.000}$), hand breadth ($0.352^{0.000}$), foot breadth ($0.315^{0.000}$), elbow-to-elbow breadth ($0.411^{0.000}$), and shoulder bideltoid breadth ($0.159^{0.000}$)), and 7 dimensions for males (body mass ($0.219^{0.004}$), shoulder height sitting ($0.166^{0.031}$), elbow height sitting ($0.192^{0.012}$), abdominal depth ($0.512^{0.000}$), hand breadth ($0.270^{0.000}$), foot breadth ($0.216^{0.005}$), and elbow-to-elbow breadth ($0.230^{0.003}$)). Particularly, age had a large effect on abdominal depth, as expected. A significant and positive correlation was found between age and BMI for both males ($0.372^{0.000}$) and females ($0.386^{0.000}$), which was also observed in Heinck (2006).

For positive correlations with age, most dimensions could also be linked to an increase in BMI in both men and women (i.e., body mass, abdominal depth, hand breadth, foot breadth, elbow-to-elbow breadth, and shoulder bideltoid breadth). On the other hand, the positive age-correlations of shoulder and elbow height sitting, which was found for men, could be explained by changes in the shape of the bony shoulder girdle (U.S Department of Health, Education, 1970). Nevertheless, it is more likely that the last two correlations were influenced by the low number of males aged 56 to 65.

4.2.2. BMI

The BMI standard weight status categories (CDC, 2021) were used to classify the BMI of males and females, as displayed in Table 6. Accordingly, 1.7% ($n = 3$) of females and 0.0% ($n = 0$) of males were 'underweight', 54.0% ($n = 94$) of females and 43.2% ($n = 73$) of males were classified as 'healthy weight', and 44.1% ($n = 73$) of females and 56.8% ($n = 96$) of males pertained to the 'overweight' or 'obesity' categories ($\text{BMI} > 25.0$). The men were 1.6 times more likely to be overweight than women, while women were twice as likely to be obese. Statistically, it was observed no significant difference between the BMI of males ($Z = 7199.000$; $p\text{-value} > 0.05$) and females ($Z = 6396.000$; $p\text{-value} > 0.05$).

When stratified by age groups (Table 7), it was observed an incidence of overweight and obesity in individuals aged 56–65 years. Comparing the different age categories, 20.0% ($n = 6$) of individuals in the 16–25 age group were overweight or obese, compared to 38.0% ($n = 39$) in the 26–35 age group, 61.3% ($n = 46$) in the 36–45 age group, 59.0% ($n = 62$) in the 46–55 age group, and 80.0% ($n = 16$) in the 56–65 age group. When stratified by gender, the observed rate of overweight and obesity in those aged 56–65 years was 100.0% in males ($n = 7$) and 69.2% in females ($n = 9$) was consistent with the results reported in other studies, such as Corish and Kennedy (2003).

Observing the BMI from the perspective of work shifts, a higher concentration of obese individuals was observed during the night shift. For men, the incidence of obesity in the daily shift was 6.0% and 15.0% in the night shift. For women, the incidence of obesity in the daily shift was 13.0% and 25.0% in the night shift. The impacts of working at night on individuals' health have been largely studied in the literature. It has been associated with less sleep hours, higher weight, BMI, and abdominal circumference (Brum et al., 2020). However, the work of Bekkers et al. (2015) could not confirm the association between shifts and weight gain. Particularly, it has been observed a higher incidence of breast cancer in women working night shifts, but only after a long period of time (Åkerstedt et al., 2015). Moreover, men ($0.411^{0.000}$) and women ($0.409^{0.000}$) in the daily shift showed a statistically significant, medium,

Table 6

Number of individuals per BMI category.

Gender	Underweight (Below 18.5)	Healthy Weight (18.5–24.9)	Overweight (25.0–29.9)	Obesity (30.0 and above)	Total
Female	3	94	53	24	174
Male	0	73	84	12	169
Total	3	167	137	36	343

Table 7

Cross tabulation between BMI and age groups by number of individuals.

Classification	Age groups (years old)					Total
	16–25	26–35	36–45	46–55	56–65	
Underweight	0	2	0	1	0	3
Healthy Weight	24	68	29	42	4	167
Overweight	6	39	36	43	13	137
Obesity	0	4	10	19	3	36
Total	30	113	75	105	20	343

and positive correlation between BMI and Age. This behaviour was also observed when removing the effect of the shift on the BMI (male, (0.372^{0.000}); female, (0.386^{0.000})).

It was also noted that some dimensions showed a positive correlation as a function of BMI, which was more expressive in males. In the female population, the BMI presented a significant positive correlation with 11 dimensions (body mass (0.878^{0.001}), elbow height sitting (0.164^{0.030}), thigh clearance (0.641^{0.000}), abdominal depth (0.826^{0.000}), hand breadth (0.192^{0.011}), foot breadth (0.234^{0.002}), elbow-to-elbow breadth (0.867^{0.000}), shoulder bideltoid breadth (0.600^{0.000}), shoulder breadth biacromial (0.176^{0.020}), hip breadth (0.649^{0.000}), age (0.382^{0.000})) and a significant negative correlation with 7 dimensions (stature (-0.318^{0.000}), eye height (-0.335^{0.000}), shoulder height (-0.300^{0.000}), elbow height (-0.241^{0.001}), sitting height (-0.237^{0.002}), eye height sitting (-0.198^{0.009}), popliteal height (-0.406^{0.000})). Stronger correlations were found between BMI and BM, elbow-to-elbow breadth, abdominal depth, hip breadth, thigh clearance, and shoulder bideltoid breadth.

Conversely, in the male population, there were 17 positive BMI correlations (body mass (0.842^{0.000}), grip reach forward reach (0.164^{0.033}), shoulder height sitting (0.253^{0.001}), elbow height sitting (0.403^{0.000}), buttock-knee length (0.169^{0.028}), thigh clearance (0.729^{0.000}), elbow grip length (0.349^{0.000}), abdominal depth (0.773^{0.000}), hand breadth (0.383^{0.000}), foot breadth (0.430^{0.000}), foot length (0.212^{0.006}), elbow-to-elbow breadth (0.768^{0.000}), shoulder bideltoid breadth (0.561^{0.000}), shoulder breadth biacromial (0.256^{0.001}), subscapular height (0.232^{0.002}), hip breadth (0.604^{0.000}), age (0.341^{0.000}) and one negative (popliteal height (-0.166^{0.032})). Stronger correlations were found between BMI and BM, abdominal depth, elbow-to-elbow breadth, thigh clearance, hip breadth, and shoulder bideltoid breadth.

For both men and women, most positive correlations with BMI can be explained by fat accumulation in some areas of the body, such as lower limbs (i.e., thigh clearance and foot breadth), upper limbs (i.e., shoulder bideltoid breadth, shoulder breadth biacromial, elbow grip length, and hand breadth), hips and buttock (hip breadth, buttock knee length), trunk and abdomen (abdominal depth, elbow to elbow breadth, and elbow height sitting). Accordingly, Chumlea et al. (2002) have observed that wider hips, arm frames, and knees are linked to increased amounts of adipose tissue and muscle throughout the body. Theoretically, women are prone to accumulate fat tissue around the hips and thighs and men around the trunk and abdomen (Bredella, 2017; Wells, 2007). However, our results did not show any difference in fat accumulation in these regions as a function of gender, as both groups displayed significant correlations between hip breadth, thigh clearance, abdominal depth, and BMI.

Amongst males, other positively correlated dimensions with BMI included foot length, subscapular height, shoulder height sitting, and grip reach forward reach. An accumulation of fat in the buttock could explain higher shoulder height sitting and subscapular height, whereas a similar process in the upper back could cause an increased grip reach forward reach length. However, subscapular height can be relatively difficult to measure, and this could have influenced the obtained correlation. Shoulder height is also subject to differences between body sides given asymmetries caused by the dominance side and postural imbalances (Christie et al., 1995; Yagi et al., 2013). Moreover, an

increased foot length, even though less correlated than foot width, can be linked to a foot flattening process from a reduction in foot arch height (Price and Nester, 2016).

Negative correlations between popliteal height and BMI can be explained by increased fat tissue in the thighs, which is evidenced by a positive correlation between thigh clearance and BMI. For women, other negatively correlated dimensions (stature, shoulder height, elbow height, sitting height and eye height sitting) could relate to postural issues caused by a higher BMI. Steele et al. (2020), for instance, showed that higher BMIs in women were directly linked to the accumulation of fat in the breasts and greater thoracic flexion torques. In obese women, this was even linked to a higher incidence of thoracic kyphosis, which increases the upper back rounding and consequently reduces vertical spine-related measures (Steele et al., 2020).

4.2.3. Other correlations

The remaining correlations between anthropometric variables, BMI, and age are in the Supplementary Material (Appendix A). For males, 77.4% of the obtained correlations were significant at p -value ≤ 0.01 , 5.0% were significant at p -value ≤ 0.05 , and the remainder 17.5% were non-significant (p -value > 0.05). For females, 64.9% of the correlation values were significant at p -value ≤ 0.01 , 9.3% were significant at p -value ≤ 0.05 , and 25.8% were non-significant (p -value > 0.05).

4.3. Comparing anthropometric databases: present study v. Barroso et al. (2005)

The data were compared with Barroso et al. (2005) to observe the differences between the current Portuguese working population and that of 17 years ago. The gender and age of the sample populations in Barroso et al. (2005) and the present study are compared in Table 8. Concerning methods, the present study comprehensively evaluated 27 anthropometric dimensions through traditional measurement and one index (BMI) (Table 1), while Barroso et al. (2005) considered 25 anthropometric dimensions measured with a photogrammetric method (2D).

Even though some authors argue that different methods can be comparable (Meunier and Yin, 2000), there may be differences when measuring anthropometric dimensions with methods of different nature, such as 2D and 3D scanners and traditional equipment. Barroso et al. (2005) compared the photogrammetric method (2D) with an anthropometer to assess the validity of the new data collection method. For the validation, the authors used the absolute value of the difference between the two measurements (i.e., the mean absolute difference (Feathers et al., 2004)). Particularly, mean absolute differences were presented for five specific anthropometric dimensions in Barroso et al. (2005), as shown in Table 9.

Therefore, given the absence of mean absolute differences for all anthropometric dimensions and genders in Barroso et al. (2005), only gender-weighted average values for the five dimensions in Table 9 were compared, as shown in Table 10.

All dimensions exceed intervals of mean absolute differences shown

Table 8

Sample population characteristics: present study v. Barroso et al. (2005).

Sample characteristics		Present study	Barroso et al. (2005)
Gender	Male population	49.3%	55.2%
	Female population	50.7%	44.8%
Age	Less than 20		
	Male	0.6%	10.2%
	Female	0.0%	8.0%
	20–50		
	Male	86.4%	81.9%
	Female	62.1%	78.5%
	More than 50		
	Male	13.0%	11.4%
	Female	37.9%	10.0%

Table 9

Mean absolute differences (in mm) between methods. Source: Barroso et al. (2005).

Dimension	Traditional method vs. Photogrammetric method
Stature	7
Knee height	2
Elbow height	2
Abdominal depth	1
Hip breadth	4

Table 10

Comparison between gender-weighted mean values (in mm).

Dimension	Present study ^a	Barroso et al. (2005) ^b
Stature	1665.7	1634.0
Knee height	527.4	504.8
Elbow height	1041.9	1011.9
Abdominal depth	242.4	262.8
Hip breadth	393.4	389.0

$$^a x = 0.493x_{\text{male}} + 0.507x_{\text{female}}$$

$$^b x = 0.552x_{\text{male}} + 0.448x_{\text{female}}$$

in Table 9, except for hip breadth. This implies that even though this consists of a comparison between different methods, the stature, knee height, and elbow height are larger in the current sample than they were 17 years ago. Conversely, abdominal depth has decreased in comparison to the Portuguese working population of 2005. These apparent changes could be related to differences in the proportion of age groups between sample populations (see Table 8): i) there were almost no participants in the 'less than 20' age group and ii) more people were 50 years old or older in the present study in comparison to Barroso et al. (2005).

Particularly, the increase in the stature of workers corresponds to findings of other works, such as Galofré-Vilà's (2018) and Hanindriyo et al.'s (2018), recognised as the secular change in global height. Since the stature increased, knee and elbow height should increase as well. On the other hand, it was not considered a difference in hip breadth given a mean absolute difference of 4 mm between studies (Table 9). Lastly, the reduction in abdominal depth can be related to changes in BMI, which are discussed below.

As the BMI was not calculated in Barroso et al. (2005), it was necessary to calculate it based on the average BM and stature measures presented in Barroso et al. (2005) (Appendix B) in order to compare the BMI between populations. The relevance of having the BMI compared across populations relies on the relationship between BMI and overall health indicators, such as nutritional status, risk factors for cardiovascular and kidney diseases, a propensity to diabetes, cancer, musculoskeletal disorders (NCD Risk Factor Collaboration (NCD-RisC), 2016) and faster rates of decline of cognitive function (Yuan et al., 2021). Table 11 presents the BMI for both male and female populations in the current study and Barroso et al. (2005).

The one-sample *t*-test showed that the average female BMI in the current study was statistically different from the average female BMI from Barroso et al. (2005) ($t(174) = -2.761, p < 0.01$). On the other hand, for males, the average BMI of the current study was similar to the average BMI of Barroso et al. (2005) ($t(169) = -0.747, p > 0.05$). The BMI variation was -3.4% for the female population and -0.8% for the male population. However, even though the current BMI was smaller

Table 11

Present study v. Barroso et al. (2005): BMI. BM is given in kg, Stature in mm, and BMI in kg/m^2 .

Study	Female			Male		
	BM	Stature	BMI	BM	Stature	BMI
Present Study	63.7	1594.7	25.2	77.9	1740.7	25.7
Barroso et al. (2005)	64.0	1565.0	26.1	74.0	1690.0	25.9

than 17 years ago, the average BMI remained above the overweight threshold ($\text{BMI} > 25$). This is in line with the trend of overweight and obesity in the general Portuguese population (Santos et al., 2014) and the global epidemic of being overweight (Cole, 2003; Gordon and Bradtmiller, 2012; Heineck, 2006; NCD Risk Factor Collaboration (NCD-RisC), 2016; Tomkinson et al., 2010).

4.4. Limitations

The first limitation of this work relates to its geographical coverage as it evaluated only one company in Northern Portugal. Secondly, even though the sample population followed the imposed statistical criteria, which made the collected data significant at the sample level, this work lacks statistical significance at the national level. Therefore, it should not be used as a normative reference for the whole Portuguese population. Third, the current study and the previous (Barroso et al., 2005) used different methods to collect anthropometric data. This study used the traditional method, while Barroso et al. (2005) used a 2D image-based method, which has become obsolete nowadays. The method chosen by Barroso et al. (2005) deeply impaired the capability of future studies to evaluate the evolution of anthropometric dimensions in the Portuguese working population. However, the comparison between the two studies employed the gender-weighted average values for the five anthropometric dimensions that had their mean absolute differences between traditional and photogrammetric methods presented in Barroso et al. (2005).

The last limitation refers to the proposed BMI comparison with the work of Barroso et al. (2005). Since the latter did not bring BMI raw data, it was necessary to calculate it using the average height (stature) and weight (body mass), which did not allow a more detailed comparison across genders and age groups. Nevertheless, despite these limitations, this work proved relevant to inform potential significant changes in Portugal's working population and underscore the relevance of updating the country's last anthropometric database. It was also possible to find agreement with the secular change in stature and overweight and obesity upward trends.

5. Conclusion

This study evaluated 27 anthropometric dimensions of 343 workers ($n = 343$) from a company in Northern Portugal. The primary aim of this study was to create a database for the company and underscore the need for an updated anthropometric database for the Portuguese working population. It was motivated by the year of the latest database (i.e., 2005) and the data collection method employed in the previous study (i.e., Barroso et al. (2005)) that impaired the comparison between new studies. Therefore, 27 dimensions were measured through a traditional method, and the BMI was calculated for a sample of 343 workers. It was observed that males had larger average dimensions than females, except for hip breadth, which was expected and already reported in other anthropometric studies. The assessment of BMI values showed that men were more likely to be overweight than women, while women were twice as likely to be obese.

Since this study and the study of Barroso et al. (2005) applied different measurement methods, only 5 anthropometric dimensions, which were validated against a traditional method in Barroso et al. (2005), could be used for comparison. This comparison showed that the Portuguese working population experienced some significant changes in 4 of the evaluated dimensions. Particularly, it was noted a higher stature in the sample population than the one reported by Barroso et al. (2005), which follows a secular change observed in other studies. Concerning the BMI, which had to be calculated using the average height (stature) and weight (BM) from Barroso et al. (2005), it was perceived a small reduction for both males' and females' BMI. However, this negative variation was statistically significant only for women. Even though lower, approximately 57.0% of the males and 44.0% of the females

continued to have a BMI above 25. These figures are in accordance with overall global trends of overweight and obesity.

To conclude, this work presents itself as useful for future studies that aim to capture the evolution of anthropometric physical dimensions in Portugal, mainly because of the reproducibility of its data collection method. In relation to a future research agenda, this work indicated that some of the current anthropometric dimensions of the Portuguese working population can be significantly different from those of 17 years ago, highlighting a potential need to develop a new normative anthropometric reference for the country. Similarly, it is recommended the collection and analysis of anthropometric data for all 18 Portuguese districts, as proposed in [Padez \(2002a,b\)](#), in order to build a database that is representative at the national level. As pointed out in this work and many others in the literature in relation to ageing, immigration, and overweight/obesity trends, it is recommended that companies invest in workers' physical, psychological, and social well-being as a means to ensure a safe and healthy work environment. Considering the high number of occupational accidents and disorders, we highlight the relevance of up-to-date anthropometric data and appropriate workplace dimensioning.

CRedit author statement

P. C. Anacleto Filho (corresponding author): Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - Original Draft, Writing - Review & Editing, Visualization. **Lincoln Silva:** Investigation, Formal analysis. **Diego Mattos:** Investigation. **Ana Pombeiro:** Investigation, Resources. **H. I. Castellucci:** Methodology, Writing - Review & Editing. **Ana Colim:** Supervision, Writing - Review &

Editing. **Paula Carneiro:** Supervision, Writing - Review & Editing. **Pedro Arezes:** Supervision, Writing - Review & Editing, Funding acquisition.

Ethical standards

This study was approved by the Research Ethics Committee of the University of Minho (No. CEICVS 137/2020).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ergon.2023.103473>.

Appendix B. Present study vs. [Barroso et al. \(2005\)](#)

[Table B1](#) and [Table B2](#) present the mean values, standard deviation (SD), P5 and P95 for the 20 anthropometric dimensions common between the present study and [Barroso et al. \(2005\)](#) for the male and female populations, respectively.

Table B.1

Male populations: Average values, SD, P5 and P95 for the 20 anthropometric dimensions selected for comparison. Values are in mm, except for BM (kg).

Anthropometric Dimensions	Present study				Barroso et al. (2005)			
	X	SD	P5	P95	x	SD	P5	P95
1 BM	76.4	10.9	63.6	96.9	74	11	56	92
2 Stature	1740.5	65.0	1642.0	1846.0	1690	76	1566	1814
3 EyeH	1641.7	65.5	1542.5	1747.5	1585	74	1463	1707
4 ShoulderH	1452.4	58.9	1357.2	1545.0	1395	65	1289	1501
5 ElbowH	1089.9	49.4	1010.0	1180.0	1050	51	966	1134
7 GRFR	715.8	35.5	655.0	782.7	730	62	627	833
8 SittingH	921.6	32.6	865.0	975.0	920	37	859	981
9 EyeHS	813.4	33.5	755.0	870.0	810	34	754	866
10 ShoulderHS	626.4	27.7	580.0	670.0	630	33	575	685
11 ElbowHS	267.0	26.0	222.2	312.7	255	30	206	304
12 BKL	601.1	27.6	560.0	645.0	590	33	536	644
13 KneeH	550.0	40.0	517.5	600.0	525	30	475	575
14 BPL	500.0	32.5	460.0	550.0	485	32	432	538
15 PoplitealH	435.0	40.0	395.0	472.7	400	26	358	442
16 ThighC	131.9	10.7	115.0	151.0	175	17	146	204
17 EGL	354.0	27.5	32.8	38.7	350	18	320	380
18 Abdominal	248.0	41.5	213.5	308.5	265	32	213	317
24 SBD	463.0	28.5	427.0	512.1	475	30	425	525
25 SBA	386.0	45.0	328.5	430.0	335	22	299	371
27 HipB	382.0	31.0	353.5	433.0	380	24	340	420

*Body mass (BM), stature (Stature), eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), grip reach forward reach (GRFR), sitting height (SittingH), eye height sitting (EyeHS), shoulder height sitting (ShoulderHS), elbow height sitting (ElbowHS), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), popliteal height (Popliteal H), thigh clearance (ThighC), elbow grip length (EGL), abdominal depth (Abdominal), shoulder bideltoid breadth (SBD), shoulder breadth biacromial (SBA), and hip breadth (HipB).

Table B.2 Female populations: Average values, SD, P5 and P95 for the 20 anthropometric dimensions selected for comparison. Values are in mm, except for BM (kg).

Anthropometric Dimensions*	Present study				Barroso et al. (2005)			
	x	SD	P5	P95	x	SD	P5	P95
1 BM	63.7	10.2	49.6	82.9	64	10	48	80
2 Stature	1593.0	63.6	1492.7	1687.5	1565	66	1456	1674
3 EyeH	1497.3	62.1	1393.7	1595.0	1465	66	1356	1574
4 ShoulderH	1320.0	82.5	1225.0	1431.2	1295	56	1203	1387
5 ElbowH	997.5	57.5	918.7	1080.0	965	46	890	1040
7 GRFR	645.9	30.0	595.0	696.5	675	33	620	730
8 SittingH	855.5	31.7	800.0	905.0	865	35	807	923
9 EyeHS	755.0	28.3	708.7	800.0	760	35	703	817
10 ShoulderHS	580.0	35.0	538.7	626.2	595	34	539	650
11 ElbowHS	258.4	24.6	220.0	296.2	250	28	204	296
12 BKL	561.6	26.0	520.0	600.0	570	32	518	622
13 KneeH	502.3	25.6	460.0	545.0	480	27	435	525
14 BPL	475.0	30.0	440.0	520.0	470	30	421	520
15 PoplitealH	387.4	25.1	350.0	430.0	365	23	327	403
16 ThighC	124.0	15.0	107.7	148.0	165	15	140	190
17 EGL	319.0	16.1	290.7	345.3	320	17	292	348
18 Abdominal	223.5	50.0	180.7	298.0	260	36	201	319
24 SBD	405.0	23.5	380.0	441.5	445	31	394	496
25 SBA	346.0	41.5	297.7	383.2	300	25	260	341
27 HipB	396.5	42.0	350.7	456.5	400	27	355	445

*Body mass (BM), stature (Stature), eye height (EyeH), shoulder height (ShoulderH), elbow height (ElbowH), grip reach forward reach (GRFR), sitting height (SittingH), eye height sitting (EyeHS), shoulder height sitting (ShoulderHS), elbow height sitting (ElbowHS), buttock-knee length (BKL), knee height (KneeH), buttock-popliteal length (BPL), popliteal height (Popliteal H), thigh clearance (ThighC), elbow grip length (EGL), abdominal depth (Abdominal), shoulder bideltoid breadth (SBD), shoulder breadth biacromial (SBA), and hip breadth (HipB).

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